

The schematic diagram illustrates the electrical connections for a custom PCB. The components and their connections are as follows:

- U1: ESP32-S3-ZERO**
 - Power:** 5V input, 3V3 (OUT) output to +3.3V rail. PWR_FLAG signal.
 - GPIOs:** GP6 (din), GP7 (right), GP8 (left), GP9 (dc), GP10 (bclk), GP11 (sda), GP12 (scl), GP13 (rst). GND connection.
 - Other:** RX, TX, RX+, RX-.
- U2: Audio Amp MAX98357A_Breakout**
 - Power:** +3.3V input, GND connection.
 - Control:** LRC (1), BCLK (2), DIN (3), GAIN (4), SD (5), GND (6), VIN (7).
- U3: Boost_Charger_18x23.6**
 - Power:** 5V input, 3.3V output to +3.3V rail. PWR_FLAG signal.
 - Output:** OUT+, OUT- connected to +5V rail.
 - Capacitors:** C9 (100uf), C10 (100nf).
- U4: TFT_Screen_2.4"1 ST7789V**
 - Power:** +3.3V input, GND connection.
 - Control:** scl (3), sda (4), rst (5), dc (6), cs (7).
- J3: PCB Board Female Header Conn_01x09_Pin**
 - Power:** +5V input, GND connection.
 - Control:** time (9), Game_b (8), game_a (7), jump (6), down (5), up (4), right (3), left (2), GND (1).
- Other Components:**
 - SW8 SW_SPDT:** Switch controlled by PWR_FLAG, connected to J4 B+ and J5 B-.
 - C6:** 100uf capacitor connected to SW8 and PWR_FLAG.
 - J7:** 5V_Connector_testing header connected to +5V and GND.

Front Panel Pin Headers
Conn_01x09_Pin
J6

Diagram showing the front panel pin headers (Conn_01x09_Pin J6) and their connections to various components. The headers are labeled with pin numbers 1 through 9, and the connections are shown as red lines with labels.

Left Header Connections:

- Pin 1: GND
- Pin 2: left
- Pin 3: right
- Pin 4: up
- Pin 5: down
- Pin 6: jump
- Pin 7: game_a
- Pin 8: game_b
- Pin 9: time

Right Header Connections (A and B):

- Pin 1: GND
- Pin 2: left
- Pin 3: right
- Pin 4: up
- Pin 5: down
- Pin 6: jump
- Pin 7: game_a
- Pin 8: game_b
- Pin 9: time

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